

# Jared Yu

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## Work Experience

### Boeing

Data Engineer

Remote

Jun 2024 – Present

- Built a pluggable data lake chatbot for self-service data analytics with integrated Wiki based on composable data types, personas, query engine, agentic search capabilities (using a custom harness), and scientific notebook explorer.
- Led a team of AI Engineers to develop end-to-end pipelines powered by agentic AI to parse unstructured data for various use cases. Mentored mid- and junior-level developers to architect and build data pipelines that can parse documents, recommend fixes to human text inputs, detect anomalous user groups based on metadata, apply embeddings to connect sparse datasets, integrating evals and guardrails, along with human-in-the-loop verification.
- Developed custom modules to specifically handle internal company APIs that are configured to handle sensitive export-controlled data via US-sovereign GPT models.
- Led a project for automating the labeling of export-controlled data, developing an app for a \$100M+ project that assists classification specialists and authorized experts to quickly categorize, label, and migrate millions of documents. Integrated various classification methods from regex to transformers to assist in classification.
- Developed a scalable data pipeline utilizing AWS Lambda to handle the embarrassingly parallel problem of many small files. This made it possible to efficiently process hundreds of gigabytes of text data.
- Collaborated with offshore teams to distribute tasks around app development.
- Led the data engineering effort for a modular data lake that leveraged AWS Lake Formation (LF) for FGAC. Pioneered as one of the first groups to test and develop the LF and Iceberg Terraform API with the platform team.
- Developed multiple end-to-end data pipelines using PySpark jobs in Glue, integrating a modular repo that could be modified and reused for new use cases. Built on a medallion architecture with encoding for non-US person access.
- Designed and architected optimal solutions to handle both business needs while incorporating the latest features of AI, Iceberg, SageMaker, Bedrock, Glue, and LF. Leveraged Terraform, Bedrock, Lake Formation, Lambda, DynamoDB, IAM, EventBridge, CloudWatch, Glue, Athena, SageMaker, Redshift, and S3 as foundational components for our team's set of established patterns.
- Foundational member of the team that developed various workflows from scratch, scaling the team from 2 to 50+ onshore and offshore developers. Documented designs, specifications, and diagrams in a project Wiki enhanced by AI.

### Apple

Data Engineer

Cupertino, CA

Apr 2023 – Jun 2024

- Designed and implemented a scalable ELT pipeline utilizing Kubernetes, S3, Iceberg, Trino, Kafka, Flink, Airflow, dbt, Spark, and Snowflake for terabytes of lab data, optimizing resource utilization by reducing compute time by 10+ hrs/week.
- Enhanced pipeline efficiency through custom logging and monitoring, facilitating rapid debugging; developed insightful Tableau dashboards for data visualization.
- Managed the maintenance and operations of lab instruments and data processing.
- Held scrum meetings for projects with ownership, keeping stakeholders up to date on progress.
- Improved data quality and cost efficiency by building an automated anomaly detection pipeline in Spark and Snowflake, saving 6+ hrs/week in manual work.
- Streamlined processes by creating a web-based maintenance dashboard in Streamlit for quick issue identification and resolution.
- Constructed a KPI dashboard and ELT pipeline using Kubernetes, Airflow, Spark, Snowflake, Plotly, and Streamlit for terabytes of user data.
- Increased user productivity through improved data observability.
- Implemented DevOps principles on all projects to ensure maintainability and ease of operations using Docker, Helm, and Pulumi.
- Enhanced user experience and data ecosystem transparency by implementing a data catalog and data lineage solution.
- Elevated data quality by establishing a data validation pipeline using the write-audit-publish pattern with Great Expectations, whylogs, and summary statistics dashboards.
- Leveraged statistical analysis (regression, clustering, dimension reduction) to identify anomalies and gain deep insights from complex user datasets.
- Built a diverse portfolio of statistical, machine learning, and deep learning models with PyTorch, TensorFlow, and scikit-learn addressing problems from time series modeling to data generation.

**Personal blog, Kaggle, Medium**  
Writer / Blogger

*Remote*  
Sep 2017 – Present

- Authored blog posts for an online community of data scientists, garnering over 25,000 page views in total.
- Developed clear and concise explanations of complex data science concepts for a broad audience.

**Pacific Gas & Electric**  
Business Analyst

*San Ramon, CA*  
Nov 2021 – Apr 2023

- Developed and implemented an ETL pipeline using Python, SQL, Power Query, DAX, M, and VBA to automate weekly and monthly KPI reports, reducing delivery time by over 80%.
- Scripted data processing tasks in Python to improve efficiency and minimize manual errors for colleagues.
- Performed data analysis and generated novel insights delivered in comprehensive reports with visualizations using Python and Power BI. Designed interactive dashboards to transform complex data into clear visuals for stakeholders.

**Waymo**  
Software Quality Operations

*Mountain View, CA*  
Mar 2021 – Nov 2021

- Designed and implemented metrics for evaluating self-driving car performance using in-house software frameworks.
- Developed SQL queries to generate scenarios for team distribution. Utilized Jira for task tracking.

## Projects

**Apache Software Foundation**  
Open Source Contributor

*Remote*  
Apr 2023 – Present

- Developed and shared a bootstrapped k8s implementation of Apache Airflow shared internally at Apple.
- Contributor to Apache Iceberg, implementing critical maintenance features for Pylceberg.
- Contributor to Apache Arrow, implementing fixes and changes for the Variant datatype.
- Contributor to Apache Fluss, implementing Python SDK features for AI/ML integration.

**Open Source Projects**  
Developer

*Remote*  
Mar 2025 – Present

- Developed ContainerClaw, a multi-agent system (MAS) and harness built around Apache Fluss. Achieves SWE-bench results comparable to harnesses from leading AI companies.
- Developed Lighthouse, a pluggable data lake chatbot that integrates a wiki and composable data primitives.

## Relevant Skills

- **Programming Languages:** Python, SQL, Java, C++, Rust, Scala, JavaScript, React, R
- **Frameworks / Libraries:**
  - **Data Engineering and Big Data:** Iceberg, Spark, Airflow, Fluss, Snowflake, Presto / Trino, StarRocks, Kafka, Flink, LakeFormation, MapReduce, Storm, ClickHouse, dbt
  - **Artificial Intelligence, Data Science, and Machine Learning:** LangChain, LlamaIndex, OpenAI, Gemini, Claude, SageMaker, Bedrock, vector databases (Pinecone, Milvus), Kiro, Antigravity, pandas, NumPy, SciPy, seaborn, matplotlib, scikit-learn, PyTorch, TensorFlow, XGBoost, Hugging Face, Optuna
  - **DevOps:** AWS, Docker, Kubernetes, Helm, GitHub, GitLab, Terraform, Pulumi
  - **Data Visualization:** Tableau, Superset, Power BI, Plotly
  - **Data Quality and Governance:** DataHub, Great Expectations, whylogs
  - **Web App Development:** Streamlit, Flask, React, TypeScript, Next.js
  - **Business Tools:** Excel, VBA, Power Query, M, DAX
- **Databases:** S3, DynamoDB, Snowflake, Hadoop, HDFS, Hive, SQL Server, RDS, Redshift, Postgres, MySQL, Cassandra, MongoDB, Neo4j, Pinecone, Milvus

## Education

**Johns Hopkins University**  
M.S. Data Science (Honors)

*Baltimore, MD*  
2021

**University of California, Davis**  
B.S. Statistics

*Davis, CA*  
2019